Ojasva Mishra

Z Fellow | Software Engineering | AI/ML | Quantitative Research | Robotics
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Education

University of Florida — Expected May 2029
B.S. Computer Engineering and Finance (dual major) | GPA: 4.0/4.0 | YC Startup School 2026 | SAT 1590, National Merit Commended — *Gainesville, FL*

Experience

AlgoGators Investment Fund — Sep 2026 – Present
Quantitative Research Analyst — *Gainesville, FL*
- Build systematic signals, backtests, and strategies for a ~$500K AUM student-run fund as one of two first-year students on the quant research team.
- Executed a crack-spread strategy with a **64% win rate across 1,200 live trades**, adjusting positions as hurricane forecasts changed during an active storm.

Twinly — Dec 2025 – Present
Co-Founder and CEO (Z Fellows cohort) — *San Francisco, CA*
- Build digital clones that model a user's writing, speech, and computer behavior, then run their work across hundreds of parallel instances.
- Shipped production macOS, Windows, and iOS apps to **1,000+ active users, including paying customers**; built the Identity Runtime, which lets clones operate apps and websites by combining accessibility APIs, browser DOM state, and multimodal vision with persistent memory.

Drift (Antler-backed) — Nov 2025 – Present
AI/Software Engineer (Founding Team) — *San Francisco, CA*
- Built core infrastructure and regression CI for a robotics AI coding workspace (**1,000+ users**) for configuring, running, and debugging simulations across ROS2, Gazebo, MuJoCo, and Isaac Sim; contributed to Drift Desktop (Tauri, React).
- Built autonomous evals for CLI coding agents with vision-model screenshot analysis; integrated World Labs' photo-to-3D scene generation and a 129-robot catalog.
- Traced a production Gazebo freeze to a UDP multicast bug and shipped the fix.

Carnegie Mellon University, Xu Lab — Jul 2025 – Present
Research Intern — *Pittsburgh, PA*
- With Prof. Min Xu and Dr. Xiaolong Wu, developed constraint-aware discrete-time PID gain optimization and risk-limiting conformal safety filtering for torque-limited robot joints; **first author on all three resulting papers**.
- Validated the controllers on a Franka Emika Panda arm and built SLAM and path-planning experiments from perception to actuation.

EV Bots — Apr 2025 – Feb 2026
AI Robotics Engineer Intern — *Chester Springs, PA*
- Built the company's first functional autonomous EV-charging robot prototype across mechanical design, electronics, perception, control, and closed-loop vehicle connection, plus an autonomous drone-charging dock; contributed to **multiple provisional patents**.

PteroLabs — Jul 2025 – Jan 2026
Software Engineering Intern — *Remote*
- Built an Unreal Engine 5 UAV simulator with PX4/ArduPilot/Betaflight SITL, 6-DOF flight dynamics, FPV, ML mission planning, and object detection, now used by the University of Pennsylvania for active research.

iPipeline — Jun 2025 – Aug 2025
Software Development Intern — *Exton, PA*
- Built automated test frameworks, AI agents, and developer tooling adopted across engineering teams for regression testing (Python, JavaScript, Docker).

Research and Publications

- **First author:** Constraint-Aware Discrete-Time PID Gain Optimization for Robotic Joint Control Under Actuator Saturation. *IEEE Trans. on Robotics (T-RO)*.
- **First author:** Bounding Detector-Accepted Feedback Authority in Torque-Limited Robot Control. *IEEE ICRA 2027*.
- **First author:** SCIG++: Risk-Limiting Conformal Safety Filtering for Attack-Resilient Robotic Joint Control Under Saturation. *IEEE/RSJ IROS 2026*.
- **UF NeuroSpine Research Team** (Dream Team Engineering, with UF Neurosurgery): build a spine CT and MRI segmentation pipeline (nnU-Net, MONAI, SimpleITK) that automates bone-quality measurements for surgical planning.

Projects and Competitions

Courtside | Gator Quant Hacks 2026, Systematic Trading track | Paper | GitHub | Video — Oct 2026
- Tested a pre-registered latency-arbitrage hypothesis on 13,084 Polymarket ATP/WTA markets and found wallets trading within 3 s of a point were **profitable after fees in all 11 out-of-sample months** (Markov fair value, deflated Sharpe, block bootstrap); own strategy paper-traded only.
- Built a causal computer-vision ball tracker that calls the point a median **162 ms before the bounce** on held-out table-tennis video with no wrong calls, at 119.9 fps.

Déjà View | SwampHacks XII, 1st in AI/ML and DeepSpace tracks | Live | GitHub | Video | Devpost — Oct 2026
- Built an app that turns photos or video of any space, real or in-game, into a walkable 3D world on Meta Quest (WebXR), the web, or Roblox.
- Recovered camera poses with COLMAP, trained a ~1M-Gaussian splat on HiPerGator in 11 min, filled gaps with VLM depth, and built 240 Hz physics for the VR and Roblox clients.

More projects and awards
- **HIVE** (YC Transpose, 1st place): built a computer-vision system that maps a physical workspace into task graphs that coordinate the people in it (PyTorch, React).
- **GlassBoxAI** (Startup in a Weekend Philly, 2nd of 200+): built a Zoom/Meet overlay that flags AI-assisted interview answers via eye tracking and voice analysis.
- **Apps:** FUSE (Bitrig Hacks, Swift); PreventaAI (App Store, 5.0 stars); Eyesight (LiDAR + CoreML obstacle detection); QuickCitePro (Chrome, 5.0 stars).
- **VEX Robotics, 3x World Championship Competitor:** captain, lead programmer, and driver; 7 seasons of C++ and Python autonomy and PID tuning.
- **Technology Student Association, 2x National Finalist:** captain for 6 of 8 national events.

Leadership

- **FusionBots, Co-founder:** robotics nonprofit for 200+ students; wrote its 120-page open-access Introduction to Robotics and Programming (SSRN).
- **American Rocketry Club, Founder and President:** grew it to 60+ members; led payload CAD, propulsion analysis (MATLAB), integration, and launches.
- **Youth Publications and Socioeconomic Forum, CTO:** lead technology for an AI literacy nonprofit with programs in 15+ countries.

Technical Skills

Languages: Python, C++, C#, Java, Rust, Swift, JavaScript, TypeScript, SQL, MATLAB
ML, data, and quant: PyTorch, ONNX Runtime, OpenCV, CoreML, MediaPipe, NumPy, pandas, SciPy, PostgreSQL, MongoDB, backtesting
AI engineering: LLM agents, VLMs, Claude Code, Codex, Cursor, agent evals
Infrastructure and frameworks: Linux, Git, Docker, AWS, GCP, Slurm HPC, React, Next.js, Node.js, Tauri, SwiftUI, gRPC, WebSockets
Robotics and embedded: ROS2, Gazebo, MuJoCo, Isaac Sim, PX4, ArduPilot, Arduino, SLAM, path planning, PID and real-time control, Fusion 360